Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/22/2004
PI Name:	Marsh, Mary E		
Project Title:	Cytoplasmic/Golgi interactions in the nanofabrication of biomineral composites - a genetic approach using coccolithophores		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	77030	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
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No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ridall, Amy (University of Texas Dental Branch at Houston) Duke, Jackie (University of Texas Dental Branch at Houston)		
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Task Description:	1. The first task is to identify macromolecular elements associated with the cytoplasm and Golgi which regulate the nucleation and growth of coccolith crystals (calcite). Our approach is to generate and isolate coccolithophore mutants using RNAi which do not express a particular macromolecule - one with a putative mineralizing function - and determine how its absence affects the structure of the biocomposite in vivo. 2. The second task is to explore the effects of nucleating agents and crystal shaping elements identified in the first task on calcite nucleation and growth in vitro - both in bulk solution and in synthetic liposomes. 3. The third task is a study of coccolithophore phase transition between calcifying and noncalcifying morphotypes. The overall goal of this project is to determine how CaCO3 formation is regulated in coccolithophores, calcifying phytoplankton responsible for the majority of the CaCO3 formed in the open ocean since the early Cretaceous. Coccolithophores impact the ocean carbon cycle and may regulate the Quaternary pCO2 flux through transitions between calcifying and noncalcifying morphotypes. An understanding of the cellular processes, genetic elements, and environmental cues which regulate the expression of calcifying versus noncalcifying phases will be essential for predicting future climate change. Regulation of CaCO3 crystal size and shape by coccolithophores is relevant to nanofabrication, as biological processes can provide inspiration for development of novel and useful materials such as molecular sieves and catalysts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The goal of this project is to understand the cellular processes which regulate the biofabrication of complex architectural forms in the Golgi complex of marine phytoplankton known as coccolithophores and ultimately to determine how these processes may be effected by gravity. The most significant aspect of this project is that it explores new biology. Over the past few decades the cytoskeleton and the Golgi complex have been extensively studied, but essentially nothing is known about interactions between cytoplasmic and Golgi elements which can explain the fabrication of regular but intricately sculpted inorganic composites known as coccoliths. The new biology relates directly to the emerging field of nanobiotechnology. The principles of composite biofabrication used by coccolithophores may serve as inspiration for new methods of materials assembly in vitro (biomimetics). In addition nanofabrication of new and useful materials in vivo may be possible by genetic control of coccolithophores.
Task Progress:	In previous studies we showed that the sulfated galacturonomannan (PS3 in Pleurochrysis and CP in Emiliania) is part of the machinery which induces the growth and development of CaCO3 (calcite) protocrystals into the elaborate and intricately shaped crystals of the mature coccoliths (the calcified scales of the coccolithophores). Mutants which do not express PS3 form only small protocrystals with simple shapes (rhombs) characteristic of inorganically formed calcite. Now in collaboration with Karen Henrikson we have shown by atomic force microscopy that the galacturonomannan also influences the fine structure of calcite grown in vitro. This occurs by the preferential interaction of the polysaccharide with the acute edges of the calcite rhomb, causing the development of growth steps of reduced angle with respect to the crystallographic c axis. This trend is likely responsible for the expression of prismatic and scalenohedra-like faces observed in mature coccolith crystals. We have shown that another very acidic calcium-binding polysaccharide PS2 is required for calcite nucleation in Pleurochrysis (order Coccolithales) but is not even synthesized in Emiliania (order Isochrysidales). Hence not all species mineralize by a PS2-dependent mechanism. We are analyzing the mineral-associated polysaccharides of other genera for PS2 synthesis to determine if the PS2 mechanism is specific to the Pleurochrysis genus, the Coccolithales order, or to coccolithophores in general with the exception of Emiliania. We have isolated both calcifying and noncalcifying morphotypes from Pleurochrysis and Emiliania clones which express varying degrees of phenotype stability. We are examining environmental variables (temperature, irradiance, etc.) which influence the rate of phase transition and morphotype stability. We plan to develop an RNA interference (RNAi) approach to determine which genes are required for phase switching and CaCO3 formation. To this end we are at work constructing a coccolithophore expression vector. We are developing protocols and coccolithophore-specific probes for fluorescence in situ hybridization (FISH) in order to identify noncalcifying coccolithophores in the ocean and determine the seasonal and geographic ratios of calcifying to noncalcifying morphotypes.
Bibliography Type:	Description: (Last Updated: 09/11/2019)
Articles in Other Journals or Periodicals	Henriksen K, Stipp SLS, Young JR, Marsh M. "Biological control of calcite crystallization: AFM investigation of coccolith polysaccharide function." Amer Mineralogist, in press, as of December 2004. , Dec-2004
Articles in Peer-reviewed Journals	Marsh ME. "Regulation of CaCO3 formation in coccolithophores." Comp Biochem Physiol B Biochem Mol Biol. 2003 Dec;136(4):743-54. Review. PubMed PMID: 14662299 , Dec-2003
Books/Book Chapters	Marsh ME. "Biomineralization in coccolithophores." in "Biomineralization. Progress in Biology, Molecular Biology, and Application." Ed. Baeuerlein, E. Wiley-VCH, Weinheim, in press as of December 2004., Dec-2004
Presentation	Marsh, M.E.; Ridall, A.L.; Duke, P.J. "Morphotype switching in the regulation of pelagic CaCO3 formation by coccolithophores." 2004 NASA Cell Science Conference, Palo Alto, CA. Feb-2004